

EE8257 Information Security



Lecture 1

Introduction to Information Security

P. S. Ranaweera

E313

Department of Electrical and Information Engineering

Faculty of Engineering

University of Ruhuna



Course Outline

- Course Content 
- Lectures :
 - Tuesday : 1.30 – 2.30 p.m.
 - Thursday : 1.30 – 2.30 p.m. (Work Shops after first two weeks)
- Evaluation
- Main Reference :
 - Mark Stamp, Information Security : Principles and Practice
- Learning materials would be uploaded to LMS



Introduction to Information Security

- CIA Concepts
- Cryptography
- Access Control
- Security Protocols
- Mathematical Concepts
 - Modular Arithmetic
 - Permutations
 - Linear Algebra
 - Elliptic Curves



Why do we need IS ?

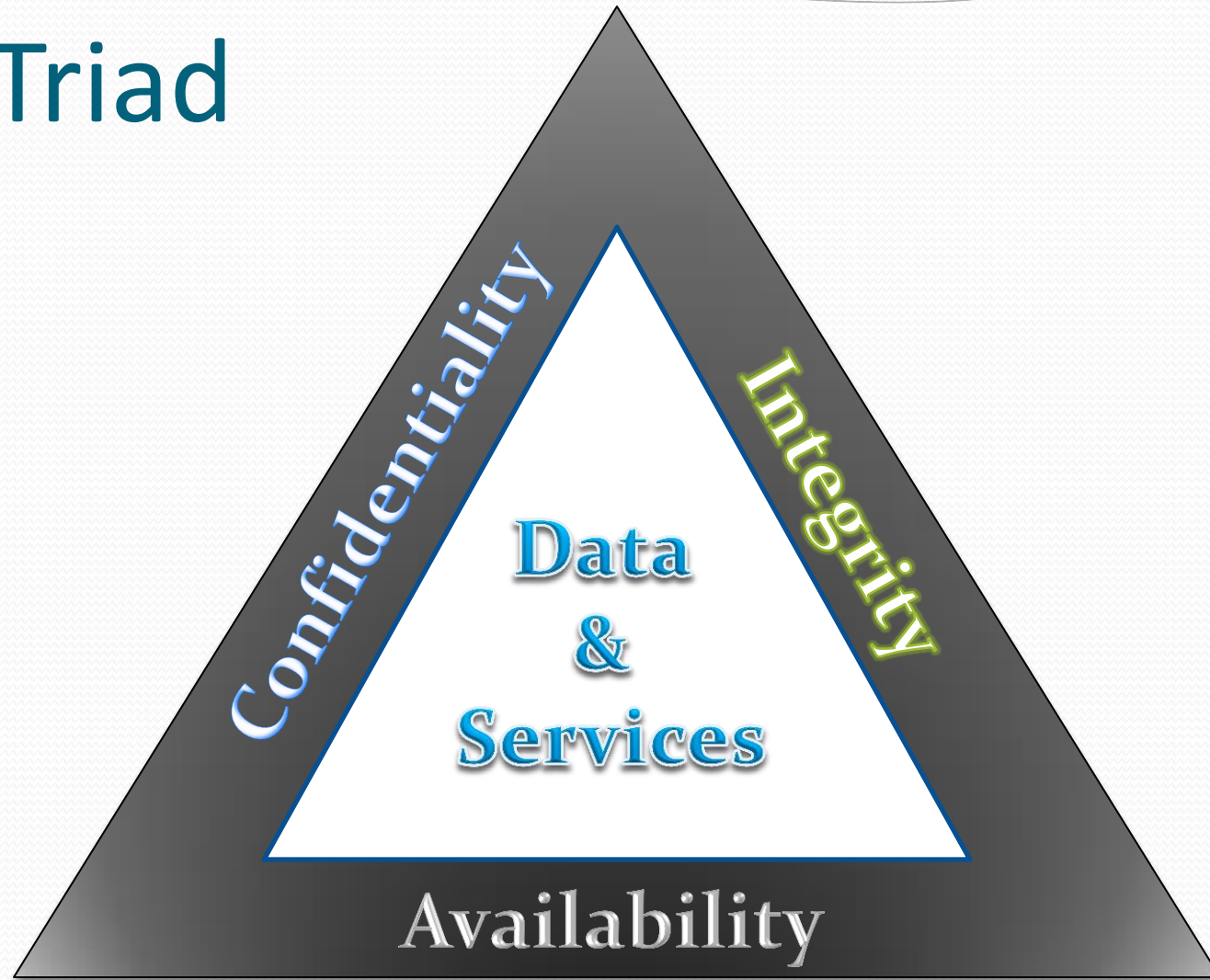
- Protect sensitive information
 - **Personal**, **Organization**, Society, State, **Country**....etc
- Prevent unauthorized manipulation of information
 - **Student**, **Employee**, **Citizen**..etc
- Protection against Malware
 - Critical Infrastructure : **PLC** → **Stuxnet**
- Prevent financial losses
 - Business deals, **biddings**, E-Commerce



Definition of IS

- Protection afforded to an automated information system in order to attain the applicable objectives of preserving the *integrity*, *availability* and *confidentiality* of information system resources [NIST].
- Resources : Hardware, Software, Firmware.....
- Storage, Processing or Transit/ Communication

CIA Triad



Key Security Concepts

- **Confidentiality** :
 - Preserving authorized restrictions on information access and disclosure
 - Hiding information to ensure privacy → Encryption
- **Integrity** :
 - Preventing information from unauthorized modification or destruction
 - Ensure non-repudiation (Denial of a committed action) ~~and authenticity~~
 - Detect modifications to prevent the misleading
- **Availability** :
 - Ensuring timely and reliable access to information

Other Security Concepts

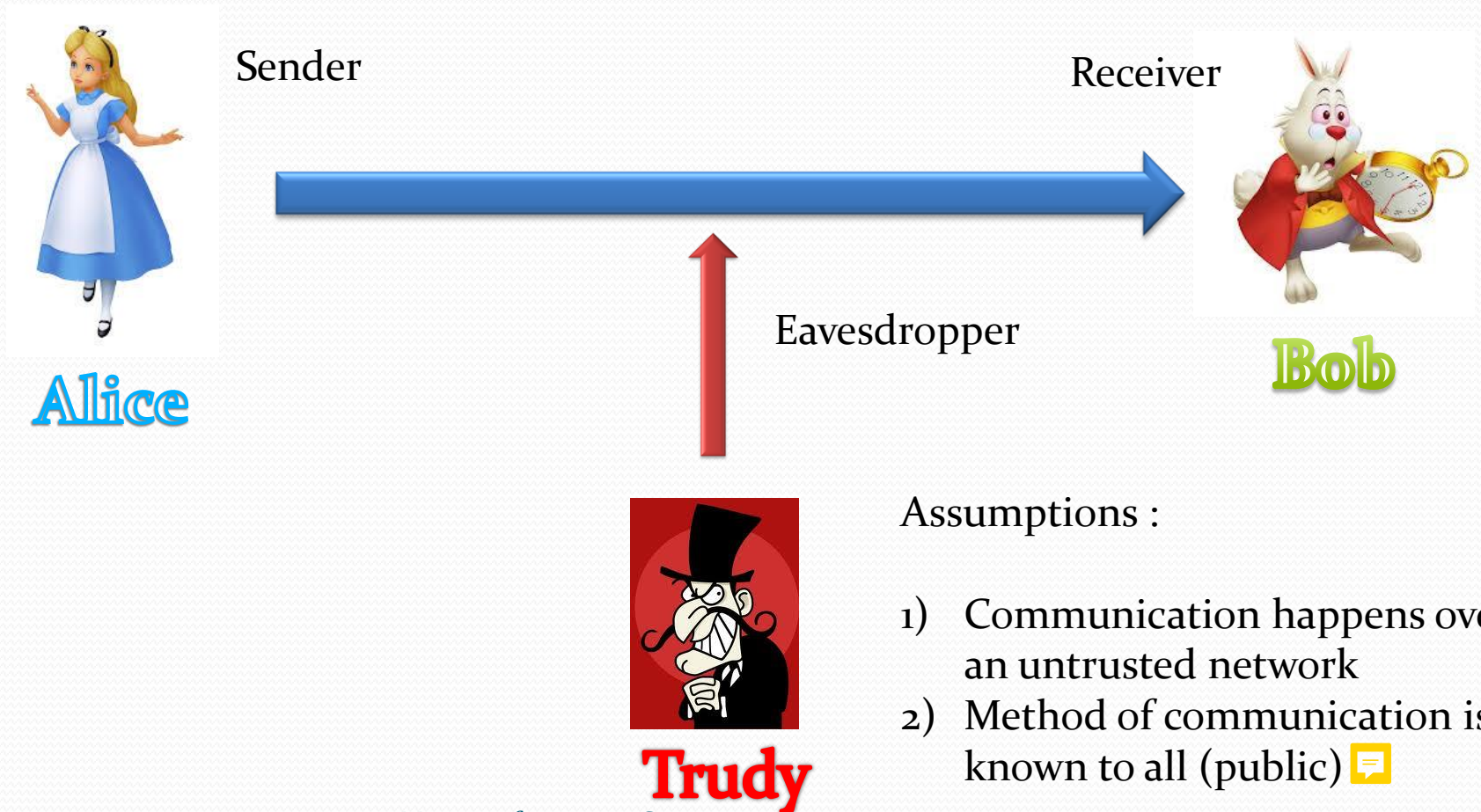
- **Authenticity** :
 - The property of being genuine and being able to be verified and trusted
 - Verifying identity through *authentication*
- **Accountability** :
 - The property that ensures the actions of an entity can be traced uniquely to that entity
 - *Identification and Traceability* of processes / actions



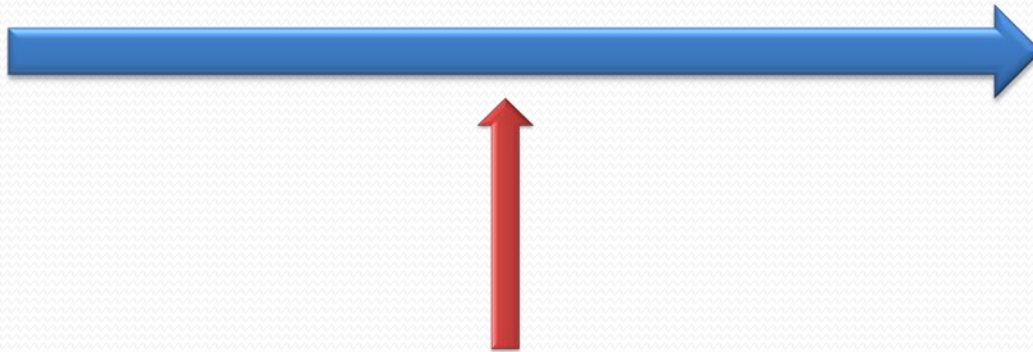
Security Terminologies

- Attacker / Hacker : An entity which poses a significant threat to the IS
- System Resource : Data, Services, Operations or Equipment
belonged to the system
- Security Policy : A set of rules and practices that specify how a system is providing security services to protect sensitive and critical resources
- Protocol : A set of rules and practices that two or more parties have established for the purpose of communication between them (an Agreement)
- Threat : A potential for violation of security policy in the system
- Vulnerability : Flaw or weakness in a system's design or operation

Security Terminology (Cont...)



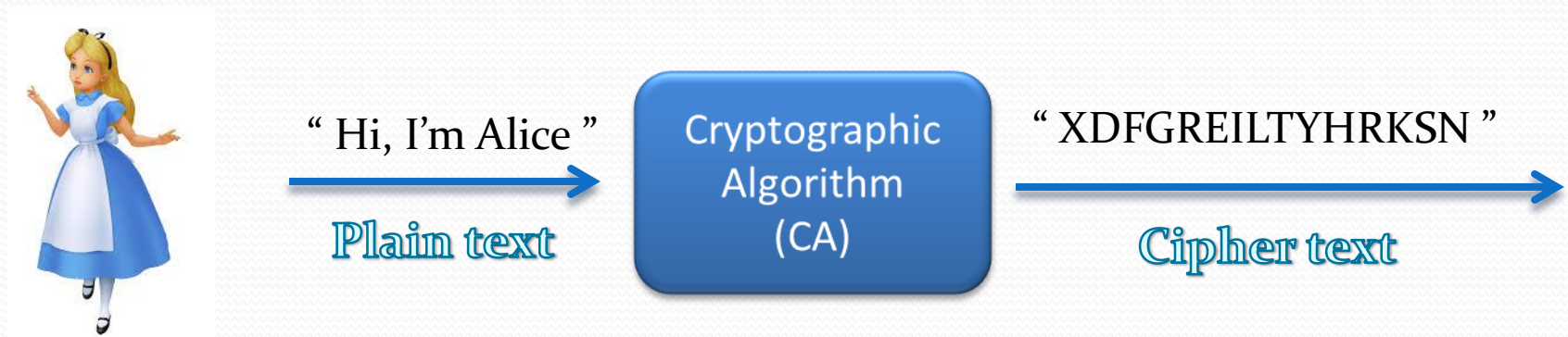
CIA with Alice, Bob & Trudy



- Confidentiality : Prevent unauthorized reading of information
- Integrity : Prevent unauthorized writing of information
- Availability : Alice and Bob both should be available

Cryptography

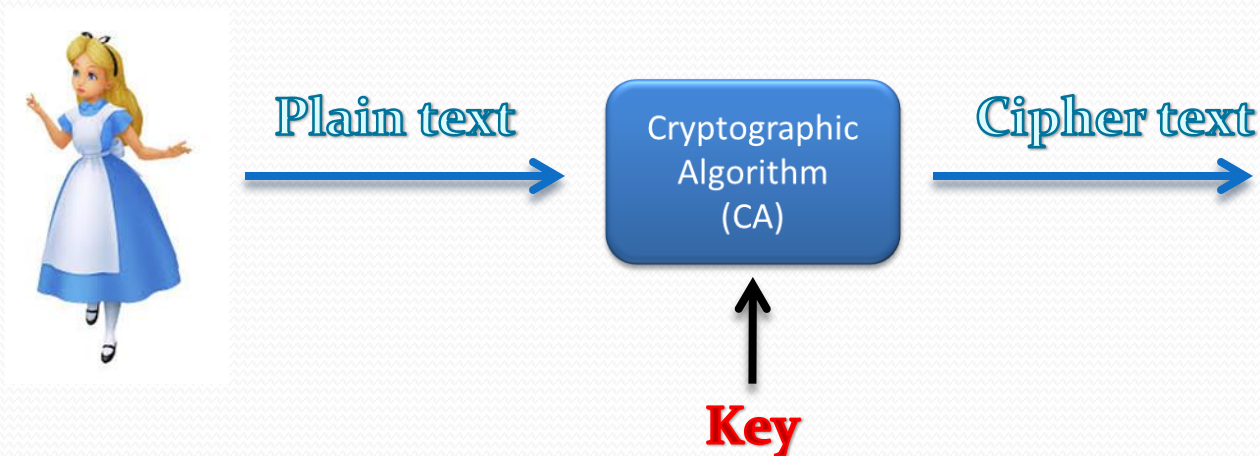
- Encrypting plaintext to produce cipher text – secret code
- Assuring Confidentiality, Privacy...



- Symmetric-Key Cryptography
- Public-Key Cryptography
- Hash Functions

Cryptographic Algorithm (CA)

- CA s are designed to achieve security goals such as CIA
- CA s rely on secrets called keys
- CA s are published – known to everyone 🗨️
- Attackers intention is to figure out or extract the key by any means possible



Access Control

- Authentication
 - Passwords
 - Biometrics
- Authorization
 - Access Control Lists (ACLs)
 - Capability Lists
 - Multi-level Security (MLS)

Log in

Don't have an account? [Create one.](#)

Username:

Password:

☐ Remember me (up to 30 days)



Security Protocols

- Simple Authentication Protocols
- Real world security protocols
 - SSL, IPSec, Kerberos,.....

Attacks

- Brute force attack
 - Exhaustive key search – try all possibilities
- Cryptanalytic attack
 - Applies mathematical techniques to obtain the key
- Non-cryptanalytic attacks
 - Does not exploit mathematical weaknesses in CAs

